

Lesson 8 - Supplement

1

Expectation As ~~the~~ Inner

Product & Norm

Def (Vector Space): A vector

space is a set V over a"field" F (in this discussion, $F = \mathbb{R}$) with two operations

$$\oplus_V : V \times V \longrightarrow V \text{ and}$$

$$\odot_V : F \times V \longrightarrow V$$

scalar product

such that:

If $u, v, w \in V$

$$1) u \oplus v = v \oplus u \quad \text{Commutativity}$$

$$2) (u \oplus v) \oplus w = u \oplus (v \oplus w) \quad \text{Associativity}$$

$$3) \exists 0_v \text{ s.t.}$$

$$u \oplus 0_v = u \quad (\text{Existence of identity element})$$

$$4) \forall v \in V, \exists (-v) \in V \text{ s.t.}$$

$$v \oplus (-v) = 0_v \quad \text{Existence of additive inverse}$$

$$5) (\alpha \odot_{\mathbb{F}} \beta) \odot_v u = \alpha \odot_v (\beta \odot_v u)$$

$$\alpha, \beta \in \mathbb{F} (= \mathbb{R})$$

(where $\odot_{\mathbb{F}}$ is the scalar multiplication)

$$6) (\alpha \oplus_{\mathbb{F}} \beta) \odot_{\mathcal{V}} u = \alpha \odot_{\mathcal{V}} u \oplus_{\mathcal{V}} \beta \odot_{\mathcal{V}} u$$

$$\forall \alpha, \beta \in \mathbb{F}$$

($\oplus_{\mathbb{F}}$ is the scalar sum.

It disappears in (6) and $\oplus_{\mathcal{V}}$

appears) (Distributivity of $\oplus_{\mathbb{F}}$
over $\odot_{\mathcal{V}}$)

$$7) \alpha \odot_{\mathcal{V}} (u \oplus_{\mathcal{V}} v) = \alpha \odot_{\mathcal{V}} u \oplus_{\mathcal{V}} \alpha \odot_{\mathcal{V}} v$$

$\forall \alpha \in \mathbb{F}$ (Distributivity of $\odot_{\mathcal{V}}$ over $\oplus_{\mathcal{V}}$)

$$8) 1_{\mathbb{F}} \odot_{\mathcal{V}} u = u$$

$$V = \mathbb{R}^{20} \quad \mathbb{F} = \mathbb{R}$$

$$x, y \in \mathbb{R}^{>0} \quad x \oplus_n y = xy$$

$$\alpha \odot x = x^\alpha$$

What is identity element $\oplus_n$

$$0_n \oplus_n u = u$$

$$0_n = 1$$

fun exercise proof above 8 axioms

Theorem: The set of all random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ is a vector space with regular sum ~~and~~ of random variables and ^{regular} scalar product $(aX, \text{ where } a \in \mathbb{R})$ ~~$\mathbb{R} = \mathbb{F} = \mathbb{R}$~~

Theorem: The set of all random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ with finite variance is a vector space with regular sum and ^{scalar} product.

L^2 space

Inner Product:

Def: Let V be a vector space over the field $F = \mathbb{R}$.

$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is called a real-valued inner product

iff:

$$1) \langle x, x \rangle \geq 0 \quad \forall x \in V$$

$$\text{and } \langle x, x \rangle = 0 \iff x = 0_V$$

(positive definiteness for inner products)

$$2) \langle y, x \rangle = \langle x, y \rangle \quad \forall x, y \in V$$

$$3) \langle x, \alpha y_1 + \beta y_2 \rangle = \alpha \langle x, y_1 \rangle + \beta \langle x, y_2 \rangle \quad \forall \alpha, \beta \in \mathbb{R}$$

Examples $V = \mathbb{R}^n$

$\langle x, y \rangle = x^T y$ is the

Euclidean inner product

$x^T Q y$ is inner product

if Q is a positive definite matrix

Def (Norm) Let $(V, \mathbb{F})$ be a vector space. We call

$\|\cdot\| : V \rightarrow \mathbb{R}$ a vector norm iff,

$$1) \|x\| \geq 0 \quad \forall x \in V$$

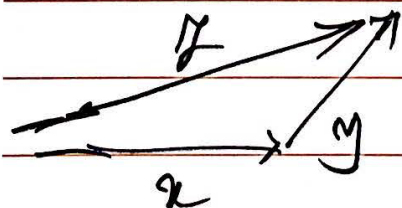
$$\text{and } \|x\| = 0 \iff x = 0_V$$

$$2) \|\alpha \odot_V x\| = |\alpha| \|x\| \quad \forall x \in V$$

$$\alpha \in \mathbb{F} = \mathbb{R}$$

$$3. \|x \oplus_V y\| \leq \|x\| + \|y\|$$

Triangle Inequality



Intuitively, the concept of a vector norm is a generalization of length in Euclidean Space.

$$x \in \mathbb{R}^n$$

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

p-norm: L^p -norm

$$x \in \mathbb{R}^n$$

$$p \in [1, \infty)$$

$$\|x\|_p = \sqrt[p]{|x_1|^p + |x_2|^p + \dots + |x_n|^p}$$

9

Norm induced by an inner product

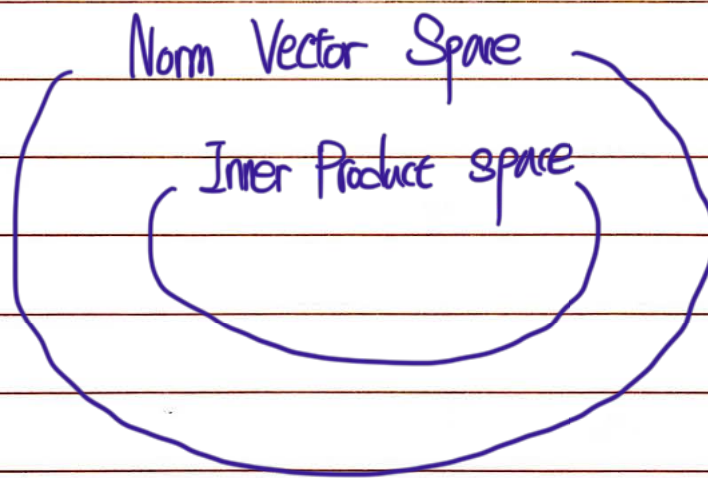
Let $\langle \cdot, \cdot \rangle$ be an inner product

on V . Then $\sqrt{\langle v, v \rangle}$ is

a norm on V and $\|v\| = \sqrt{\langle v, v \rangle}$

is called the norm induced

by the inner product $\langle \cdot, \cdot \rangle$.



Lemma (Cauchy-Schwarz-Bunyakovsky)

For a vector space V with inner product $\langle \cdot, \cdot \rangle$

$$|\langle u, v \rangle|^2 \leq \underbrace{\langle u, u \rangle}_{\|u\|^2} \cdot \underbrace{\langle v, v \rangle}_{\|v\|^2}$$

or

$$|\langle u, v \rangle| \leq \|u\| \|v\| \quad \begin{array}{l} u, v \text{ linearly} \\ \text{dependent} \end{array}$$

Equality occurs when $u = av$, $a \in \mathbb{R}$

Proof: $0 \leq \left\| u - \frac{\langle u, v \rangle}{\|v\|^2} v \right\|^2$

Lemma

$$\begin{aligned} & \langle \alpha u_1 + \beta u_2, \gamma v_1 + \delta v_2 \rangle \\ = & \alpha \gamma \langle u_1, v_1 \rangle + \alpha \delta \langle u_1, v_2 \rangle + \beta \gamma \langle u_2, v_1 \rangle + \beta \delta \langle u_2, v_2 \rangle \end{aligned}$$

Proof of CSB

$$\| u - \frac{\langle u, v \rangle}{\|v\|^2} v \| \geq 0$$

$$= \langle u - \frac{\langle u, v \rangle}{\|v\|^2} v, u - \frac{\langle u, v \rangle}{\|v\|^2} v \rangle$$

$$\begin{aligned} = & \langle u, u \rangle - \frac{\langle u, v \rangle}{\|v\|^2} \langle v, u \rangle - \frac{\langle u, v \rangle}{\|v\|^2} \langle u, v \rangle \\ & + \left(\frac{\langle u, v \rangle}{\|v\|^2} \right)^2 \langle v, v \rangle \end{aligned}$$

$$= \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2} - \frac{|\langle u, v \rangle|^2}{\|v\|^2} + \frac{|\langle u, v \rangle|^2}{\|v\|^2}$$

$$\therefore \|u\|^2 - \frac{|\langle u, v \rangle|^2}{\|v\|^2} \geq 0$$

Try to prove

$$|\langle u, v \rangle| \leq \|u\| \|v\|$$

Q.E.D.

$$\langle x, y \rangle = \mathbb{E}[xy]$$

Continuous function $f: [0,1] \rightarrow [0,1]$

$$\int_0^1 |f(t)|^2 dt < \infty$$

$$\langle f, g \rangle = \int_0^1 f(t)g(t) dt$$

CSB

$$\left| \int_0^1 f(t)g(t) dt \right|^2 \leq \int_0^1 |f(t)|^2 dt + \int_0^1 |g(t)|^2 dt$$

One can see that on the vector space of random variables with finite variance, the following inner product can be defined :

$$\langle X, Y \rangle = \mathbb{E}[XY]$$

The norm induced by the inner product is

$$\|X\| = \sqrt{\mathbb{E}[X^2]}$$

(Show that they are indeed a norm and an inner product)

Also, the Cauchy-Schwartz-Bunyakovsky inequality holds in this vector space, i.e.

$$|E[XY]| \leq \sqrt{E[X^2]} \sqrt{E[Y^2]}$$

$$\text{or } (E[XY])^2 \leq E[X^2] E[Y^2]$$

Exercise (Rewrite the proof using $E[XY]$ as the inner product)

Remark: Correlation Coefficient
is a "measure" of (linear)
dependence of X and Y .

In a Euclidean space,
such a measure is called

the directional cosine
between two vectors

$$\cos(\phi) = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

$\cos \phi = 1$ when $u = av$, $a \geq 0$

(i.e. u and v are collinear)

and have the same direction)

and $\cos \phi = -1$ iff u and v

are collinear and have

opposite directions.

Similarly, we saw that

$P(X, Y) = 1$ iff $Y = aX$, $a > 0$

and

$P(X, Y) = -1$ iff $Y = aX$, $a < 0$

Remark: Our previous discussion shows the relationship between linear algebra and probability theory (obviously, they are related in many other ways)

The set of all random variables can be viewed as a vector space. If one is interested in defining "length" and "direction" for random

variables, one needs to

Consider the subspace of
a random variables with finite variance.
(subset of a vector space that

is a vector space itself is
called a subspace)

The square root of the second
moment of a random variable

in this vector space can be

seen as its length and

$E[XY]$ can be seen as

the inner product between

X and Y . Also, $\frac{E[XY]}{\sqrt{E[X^2]}\sqrt{E[Y^2]}}$

$= \frac{\langle X, Y \rangle}{\|X\| \|Y\|}$ can be viewed as

when X
 Y have
zero mean

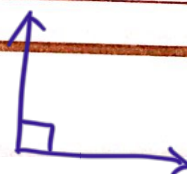
a measure of collinearity

between X and Y .

$E[XY]$ is sometimes called
 the correlation between X and Y ,
 $\text{Corr}(X, Y)$
 Obviously, $\text{Cor}(X, Y) = \text{Corr}(\tilde{X}, \tilde{Y})$.

Orthogonality

In vector spaces that are
 equipped with an inner product,
 orthogonality is defined as
 having "zero collinearity".



This is an extension of the concept of orthogonality in the Euclidean Space.

Def: Assume V is a vector space equipped with

inner product $\langle u, v \rangle$

u, v are called "orthogonal"

iff $\langle u, v \rangle = 0$.

Therefore, the concept of having zero correlation is analogous to the concept of orthogonality.

Def: Assume that X, Y are L^2 and zero-mean r.v.'s on $(\Omega, \mathcal{F}, \mathbb{P})$. They are

called orthogonal if $\mathbb{E}[XY] = 0$

(they have zero correlation)

Remark: By L^r r.v.'s, we

mean r.v.'s whose r^{th} moments

are finite.

Remark: All of the above discussions about norm, inner product, Cauchy-Schwartz inequality etc can be stated for random variables or "centralized

random variables." In the

latter case, $\|\tilde{X}\| = \sqrt{\text{Var}(X)} = \sigma_X$

$$\langle \tilde{X}, \tilde{Y} \rangle = \text{Cor}(X, Y),$$

$$\frac{\langle \tilde{X}, \tilde{Y} \rangle}{\|\tilde{X}\| \|\tilde{Y}\|} = \rho(X, Y)$$